Calum Smith

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EDUCATION

University of Warwick (Classification Pending)
MSc Statistics with Data Science Sep 2025 – Sep 2026

Modules: Monte Carlo Methods, Statistical Inference, Bayesian Forecasting, Machine Learning

Durham University, St. Chad's College
*MSc Finance — **Distinction**; Business School Scholarship* Oct 2019 – Sep 2020

University of Manchester
*BSc (Hons) in Mathematics — **2:1*** Sep 2015 – Jul 2018

RESEARCH & AI SAFETY

A Practical Block-Sampling Particle Filter: Amortising Online Inference 2026
Warwick MSc dissertation, supervised by Prof. Adam Johansen

- Extended block-sampling particle filters beyond linear Gaussian models: a conditional VAE learns the proposal offline, so each online step costs one forward pass, and a single network serves as both proposal and artificial distribution.
- Reduced resampling in a linear Gaussian model from ~ 113 to ~ 20 steps per 500-step run as block length grew from 1 to 6, versus no reduction for an independently trained network; gains were smaller on harder models.
- Showed the ELBO does not control importance-weight variance (a χ^2-discrepancy that can be infinite while the ELBO is small), so proposals were validated through filter diagnostics, not the training loss.

AI Safety Fellowship, Technical Track | *Warwick EA & AI Societies* Autumn 2025

- Participated in the technical track; presented on variational autoencoders, the model class central to my dissertation.

SELECTED PROJECTS

Statistical Consulting: Central England Temperature | *Modelling Lead, 6-person team (Warwick)* 2026

- Led the modelling for a client brief on re-baselining temperature thresholds: built a reproducible R pipeline fitting SARIMA, Kalman state-space (KFAS) and Bayesian structural time-series (`bsts`) models to HadCET, cross-checked on HadUKGrid.
- Rejected a better-fitting trend-augmented BSTS model as its 30-year 95% intervals widened from ~ 5 to $\sim 22\,^{\circ}$C and held-out accuracy on the mean worsened; recommended the stable no-trend models.
- Built the baseline-comparison and threshold-exceedance analyses behind the recommendation to retire pre-1990 baselines (all four seasons significantly warmer, $+0.46$ to $+1.68\,^{\circ}$C); reviewed teammates' pull requests and ran reproducibility checks.

Monte Carlo Methods Coursework | *R, Python (Warwick)* 2025

- Compared random-walk Metropolis and MALA on Gaussian targets in up to 100 dimensions, deriving the MALA acceptance ratio and tuning against optimal-scaling theory; at $p = 100$, MALA's mean squared jump distance was $\sim 30\times$ larger.
- Implemented Gibbs and Metropolis samplers for the 2D Ising model, showing multimodality and poor mixing at high coupling.
- Implemented inversion, rejection and importance samplers, including an optimised t proposal (74.8% acceptance vs. 74.9% predicted) and an analysis of when importance-sampling variance is infinite.

Deep Reinforcement Learning Coursework | *Advanced Topics in Statistics & Probability* 2026

- Compared Q-learning and SARSA on GridWorld, and DQN and REINFORCE on CartPole, with a target-network ablation.

WORK EXPERIENCE

Self-Directed Mathematical Study *(concurrent: Clinical Support Worker, NHS, 10 months)* Jan 2024 – Sep 2025

- Rebuilt foundations in statistical inference and Bayesian statistics ahead of the Warwick MSc.

Securing Energy for Europe (SEFE), Manchester & London
Quantitative Analyst, Gas Pricing (Jul 2022 – Dec 2023); *Flexible Portfolio Analyst* (May 2021 – Jul 2022)

- Produced and quality-controlled daily NBP/TTF forward gas curves, the primary pricing reference for UK/Dutch B2B contracts.
- Maintained the pricing framework (intraday volatility, risk adders) and tested automated adder generation via Trayport.
- Produced monthly Cost of Power reports and reconciled data supporting £100m+ monthly billing on a flexible gas portfolio.

TECHNICAL SKILLS

- **Programming:** Python (NumPy, Pandas, scikit-learn, PyTorch), R (`KFAS`, `bsts`, `forecast`, `renv`), Git, LaTeX
- **Methods:** Monte Carlo/MCMC/SMC; Bayesian inference; state-space and time-series models; VAEs